CASE REPORT

Pseudohypoparathyroidism - A Case Report

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INTRODUCTION

Pseuduohypoparathyroidism is a rare, mostly hereditary disorder, characterized by end organ resistance to parathyroid hormone action. This condition is characterized by hypocalcemia and hyperphosphatemia, with associated dysmorphic features in the form of short stature, shortened fourth metacarpals, etc. A familial case of pseudohypoparathyroidism is being presented, where 4 siblings of a single family are affected by this disorder.

CASE REPORT

Our patients are 4 siblings-- Zeetta, 28yrs; her sisters Santha,22yrs and Treesa,18yrs; and her brother Varghese,16yrs. They hail from a village in Thiruvananthapuram district of Kerala state. It was Zeetta who first presented to us and others were subsequently evaluated. So we will describe the case of Zeetta in detail. Zeetta was admitted to our ward with one episode of generalized tonic clonic seizure. She had a history of seizure disorder for the last one year and was on treatment with phenobarbitone but was poorly controlled as evidenced by 4 recurrences during that period. She also had history of leg cramps for many years. She was born out of a non consanguineous marriage by a full term normal delivery. She had attained her developmental milestones normally. But she was growth retarded and mentally retarded thereafter. She had delayed puberty with attainment of menarche at 19 yrs of age and poorly developed secondary sexual characters.

On examination, she was found to be short statured with a height of 129 cms. Her weight was 34 kg with a body mass index of 20. The upper segment to lower segment ratio was normal. She had round facies and malformed teeth. Also there was shortening of most of the metacarpals bilaterally. No thyroid swelling was palpable. Her I.Q. was formally tested and was found to be 55(mild mental retardation). Chvostek's and Trousseau's signs for hypocalcemia were positive. Examination of systems was within normal limits.

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We proceeded with laboratory investigations. Hemogram and urinalysis were normal. Renal and liver function tests were normal. Serum calcium was 5.3 mg/dl and phosphate was 5.7 mg/dl, both were confirmed by repeating. Serum magnesium was 1.9 mg/dl and albumin was 4.2 g/dl. Serum parathyroid hormone was estimated and was 39 pg/ml (10-53 was the reference value). X-ray of hands and feet showed shortening of 1st, 3rd, 4th and 5th bilaterally and 1stmetatarsals bilaterally. Generalized osteoporosis and soft tissue calcification were also evident. CT head showed diffuse calcification of bilateral basal ganglia, thalamus and cerebellum. EEG showed mild non- specific dysfunction. Thyroid function tests showed normal T3 and T4 with raised TSH of 10μIU/ml. Chest X-ray and ECG were normal. USS abdomen showed normal ovaries.

We went on to examine the 3 younger siblings of our patient and found that all 3 were dwarf with round facies and had short metacarpals and metatarsals. All had positive Chvostek's and Trousseau's sign. The youngest sibling also had history of seizure and was on antiepileptics. Others were seizure free. All three had low serum calcium and high serum phosphate levels. X-ray of the hands and feet confirmed short metacarpals and metatarsals. They also had normal serum parathyroid levels.

So our patients are hypocalcemic and hyperphosphatemic with normal parathyroid hormone levels and features of Albright's Hereditary Osteodystrophy. So they have got pseudohypoparathyroidism type Ia.

DISCUSSION

Pseudohypoparathyroidism is a disorder characterized by hypocalcemia and hyperphosphatemia in the presence of normal or high parathyroid hormone level, and in association with skeletal dysmorphic features of Albright's Hereditary Osteodystrophy (AHO). It has got a prevalence of 3 per million with a female to male ratio of 2:1. AHO consists of short stature, round facies, shortened 4th metacarpals, obesity and subcutaneous calcifications.

Majority of cases are hereditary and a minority are sporadic. The inheritance of pseudohypoparathyroidism is explained by a process called genomic imprinting, which essentially means selective inactivation of one of paternal & maternal alleles. Defective gene is GNAS-1 gene in chromosome 20q1-3, encoding

Table 1 : Classification of Pseudohypoparathyroidism

Type	Serum calcium	Serum phosphate	Serum PTH	Urinary cAMP Response to PTH	Albright's hereditary osteodystrophy	Other hormonal resistance
PHP type Ia	Low	High	High	Low	Yes	Yes
PHP type Ib	Low	High	High	Low	No	No
PHP type Ic	Low	High	High	Low	Yes	Yes
PHP type II	Low	High	High	Normal	No	No
PPHP	Normal	Normal	Normal	Normal	Yes	No

PHP=pseudo hypo parathyroidism; PPHP=pseudo pseudo hypo parathyroidism



Fig. 1 : The three siblings. On extreme right is Zeetta



Fig. 2 : Note the short 4th metacarpal of the patient's hand shown on the right. On the left is the mother's hand shown for comparison.



Fig. 3 : Note the short metatarsals of the patient's feet.



Fig. 4 : X-ray hand showing short 1st,3rd,4th and 5th metacarpals bilaterally.



Fig. 5 : X-ray feet showing short 1st metatarsal bilaterally.



Fig. 6 : CT head showing bilateral basal ganglia calcification.

Gsα subunit of Gprotein. In renal cortex, only the maternal allele of the gene is expressed, whereas in other sites, maternal or paternal allele is expressed randomly. So if maternal allele is defective, i.e., if the disease is transmitted from mother, there will be hypercalcemia & hyperphosphatemia (i.e. , pseudohypoparathyroidism)since renal cortex is involved. But if the disease is transmitted from father, only Albright's hereditary osteodystrophy will be evident without any hypocalcemia or hyperphosphatemia (i.e., pseudo-pseudo hypoparathyroidism).

Classification of pseudohypoparathyroidism is given in the table 1.

Treatment of pseudohypoparathyroidism : Supplementation of calcium in a dose of 1 to2 gms /day and vitamin D3 0.25 to 1 micrograms/day. All four patients in our case were put on calcium &Vitamin D3 supplementation and their serum calcium levels have returned to near normal after 3 months of treatment. There was no recurrence of seizures in the patients.

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